



COW HEALTH CARE PREDICTION USING CNN MODEL

ABSTRACT

This project presents an AI-based cow health care prediction system using a Convolutional Neural Network (CNN) to detect Lumpy Skin Disease (LSD) at an early stage. Traditional monitoring methods are time-consuming, subjective, and often miss early signs of disease, leading to economic losses due to reduced milk production and poor breeding efficiency. The system allows farmers to upload cow images through a web-based application, which provides instant classification as Healthy or Lumpy along with treatment and preventive recommendations. This approach minimizes manual inspection, reduces veterinary costs, improves livestock management, and ensures timely intervention to protect the herd.

INTRODUCTION

Livestock health monitoring is vital for maintaining farm productivity and preventing economic losses. Traditional methods rely on manual inspection by farmers and periodic veterinary visits, which are time-consuming, subjective, and often fail to detect early signs of diseases. Lumpy Skin Disease (LSD) is a rapidly spreading infection that reduces milk production and affects breeding efficiency, leading to significant financial loss. This project introduces an AI-based solution using a Convolutional Neural Network (CNN) to analyze cow images and classify them as Healthy or Lumpy. The web-based system provides real-time health assessments, treatment guidance, and preventive measures, making it cost-effective, accurate, and user-friendly for farmers.

SYSTEM SPECIFICATION

Hardware Requirements

- Processor: Intel Core i3 / i5 or above
- RAM: 8 GB (4 GB minimum)
- Storage: 500 GB HDD / 256 GB SSD
- System Type: 64-bit Computer
- Display: Standard Monitor
- Input Devices: Keyboard and Mouse
- Internet Connection: Required for web access

SYSTEM SPECIFICATION

Software Requirements

- Operating System: Windows 10 / Linux
- Programming Language: Python 3.11 or above
- Deep Learning Framework: TensorFlow, Keras
- Web Framework: Flask
- Image Processing Libraries: OpenCV
- Supporting Libraries: NumPy, Pandas
- Web Browser: Google Chrome / Edge

SOFTWARE FEATURES

- User-friendly web interface
- Image upload and processing support
- CNN-based disease prediction
- Accurate classification (Healthy / Lumpy)
- Fast and real-time result generation
- Automated health analysis
- Treatment and prevention recommendations
- Secure and reliable system operation

ENVIRONMENT SETUP

- Install Python 3.9 or above
- Set up virtual environment
- Install required libraries (TensorFlow, Keras, Flask, OpenCV, NumPy, Pandas)
- Configure Flask web application
- Prepare and organize dataset
- Train CNN model
- Run the application on local server

EXISTING SYSTEM

- Manual cow health monitoring
- Visual inspection by farmers
- Periodic veterinary visits
- Time-consuming and costly
- Late disease detection

DISADVANTAGES OF EXISTING SYSTEM

- Late detection of diseases
- Labor-intensive and time-consuming
- High veterinary consultation costs
- Limited access in rural areas

PROPOSED SYSTEM

- Early detection of Lumpy Skin Disease using AI
- Image-based classification (Healthy / Lumpy)
- Real-time results via web application
- Provides treatment and preventive guidance

ADVANTAGES OF PROPOSED SYSTEM

- Early detection of diseases
- Reduces veterinary costs
- Scalable for large herds
- User-friendly web interface

